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Hyrtl's Catfish *Neosilurus hyrtlii* from Olkola Country, southern Cape York, Qld.

Photo: Michael Hammer

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The discovery of *Melanotaenia garylangei* – The Golden Rainbowfish from Dekai

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On our 2010 trip to West Papua we (Dan Dority and the authors) made our “base camp” in Sentani; we then asked the mission airlines about possible flights to remote places with available free seats. Through them we found a flight to a village called Dekai, which is located on the southern side of West Papua, in the foothills of the Brazza River – part of the Eilanden River system. The flight was during perfect weather conditions, giving us marvelous sights of Lake Sentani immediately after takeoff, followed by the Tami River system, then the mountain range and afterwards the river systems flowing to the south. We learned that Dekai was a small village with a swampy grass airstrip until just a few years ago; but now the government had decided to build up a *transmigrasi* settlement there which included infrastructure measures such as a concrete runway that suits larger airplanes. At the airport we met a friend of Dan named Akso, who had borrowed a car for us. It was a 2WD pickup car, but it worked fairly well.

Two roads were built in Dekai in order to make room for settlement: one to the west and one to the north. We decided to drive along the western road first, come back and then drive the northern road on the second day. Driving along the western road, many plantations of banana, yams and manioc could be seen. In 2010 we saw no oil palm plantations; I wonder if it is still the same



Proud fishermen Johannes and Dan with their new rainbowfish.

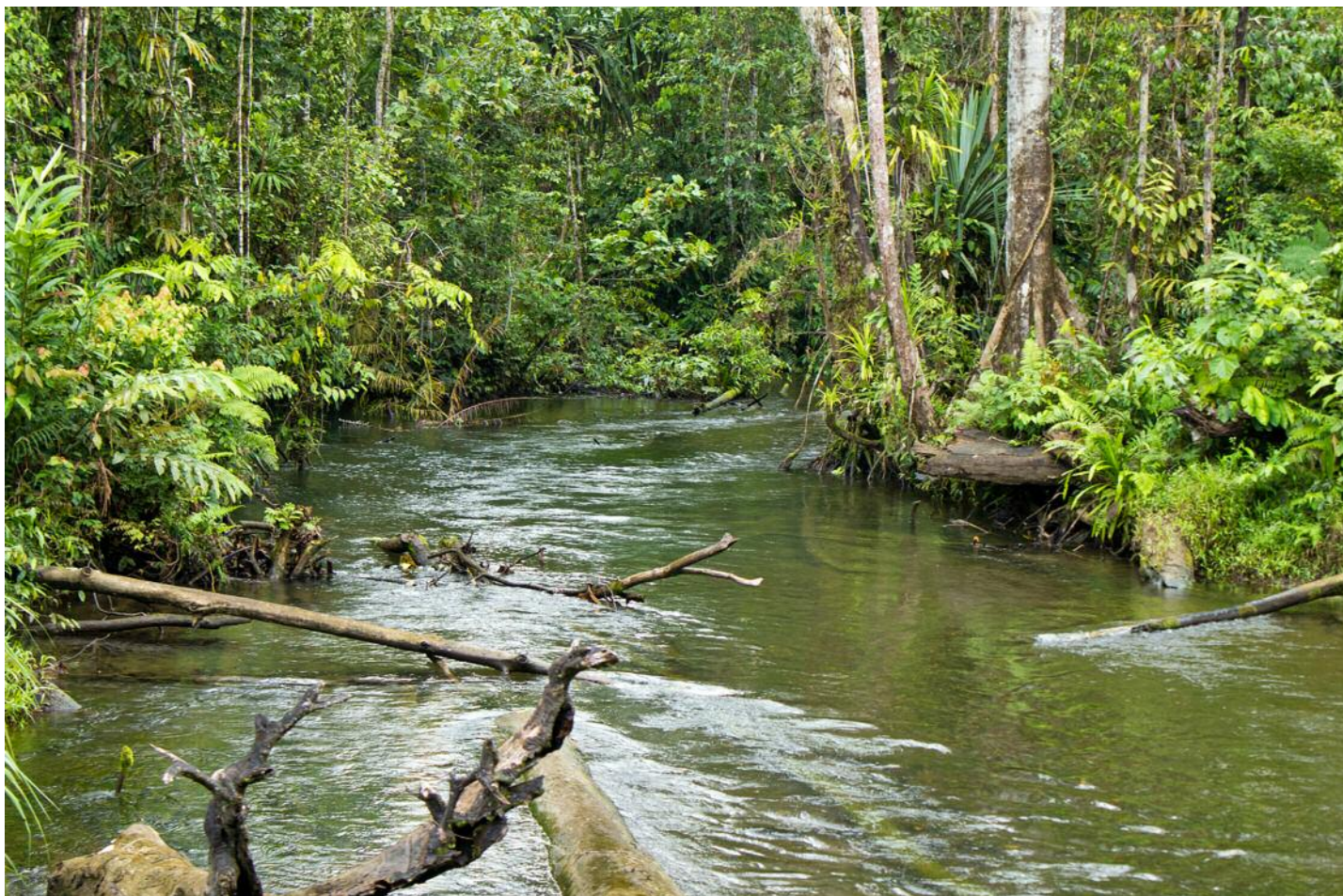
Photo: Gary Lange

today? There is a lot of rain in Dekai, about 3.8 m per year. That is a lot! Being hit by a tropical downpour while sitting in the back of a pickup car is really hard.

After a 30 minute drive, the first nice river came up – a fast flowing creek with tea-stained brown water. The first fishing site on our Dekai trip yielded a real beauty; a blue shining form of *Melanotaenia goldiei* (Goldie River Rainbowfish). This Dekai *M. goldiei* has the



Map of the Dekai region in West Papua. Base map data provided by the following sources: Google, SIO, NOAA, U.S. Navy, NGA and GEBCO



Stream typical of many that we saw in the Dekai area.

Photo: Gary Lange

brightest blue that I have ever seen so far. It also grows large: I have a male about 12 cm swimming in my living room tank. Other fishes that we found were *Zenarchopterus novaeguineae* (Fly River Garfish) and *Glossamia aprion* (Mouth Almighty). We continued our drive and came to a wider (about 10 m) and deeper (about 2 m) river. Fishing there, we found a pretty little *Craterocephalus nouhuysi* (Mountain Hardyhead) in our seine, and later on we found *Melanotaenia rubrostriata* (Red-striped Rainbowfish) near the banks where they were hiding amongst roots. Just before dark we found an interesting fishing place, but unfortunately as it was already very late, it left us very little collecting time. A quick scoop yielded small *M. rubrostriata*, *Mogurnda*



Tree chopper man.

Photo: Gary Lange



Asko, our driver.

Photo: Gary Lange



Male *Melanotaenia goldiei* from Dekai.

Photo: Gary Lange



Craterocephalus nouhuysi.

Photo: Gary Lange



Male *Melanotaenia rubrostriata*.

Photo: Gary Lange

cingulata (Banded Mogurnda) and an unknown rain-bowfish. We decided to come back the next day.

We headed back to Dekai and arrived in the dark. Akso lives in a settlement close to Dekai in a fairly new house that was built by the government. Most of the houses are surrounded by ponds where the people raise *Channa* spp. (Snakeheads) and *Oreochromis mossambicus* (Tilapia). From the local people's view it is easy to understand why they produce these fishes: they are much more tasty and not as spiny as the native species, and they are easy to keep and breed. Looking at the high rainfalls in this region it is no wonder that some fish escape and establish in natural waters adjacent to the settlements. We had a good meal with rice, fish and chicken, sitting around to chat a bit and went to sleep after a long day.

The next day we headed along the northern road, with the plan to drive to the end and then fish on the way back. After about an hours drive we were halted by a fallen tree blocking the road. Akso thought he could drive around it and promptly got stuck in the soft ground at the edge of the road! From the middle of nowhere people suddenly appeared and one of them went home to get an axe. He then chopped off the treetop while we were loading heavy stuff on the car's boot to put weight on the back wheels. With a lot of pushing we got the car out and now we had to take all the people with us for a free ride! The funny thing was the road's end was only a few hundred meters ahead! At the end of the road was a wider river. No bridge was constructed yet, but the road-building continued on the other side. We fished in the river and found some *M. goldiei*. On the other side we found a roadside ditch, heavily covered with plants. Fishing



Zenarchopterus novaeguineae, photographed in a field tank.

Photo: Gary Lange



Male *Mogurnda cingulata*.

Photo: Rachael Remington

there, we found an abundance of medium-sized *Mogurnda cingulata*, *Melanotaenia rubrostriata* and *Ambassis agrammus* (Sailfin Glassfish). The local people collected all the fishes for food.

Driving back we collected at several places, keeping our focus on small creeks. That is really hard work because the seine or frame net has to be dragged through the aquatic plants, but it was worth the effort. We found a few specimens of the new rainbowfish species that was later described as *Melanotaenia garylangei* (Golden Rainbowfish). Not many were caught, but it was the same fish as at the place the day before (on the western road), so we decided to drive to this place again. It

turned out to be a real hotspot: after a few scoops we had three species of rainbowfishes and a blue-eye, sometimes catching all four species simultaneously in one haul! The blue-eye was an unknown species of *Pseudomugil*. At this place, the four different species and their microhabitats came close together:

M. goldiei in flowing water

M. rubrostriata near quiet banks under vegetation

M. garylangei in shallow creeks of lesser depth with black water

Pseudomugil species "Dekai Village" in very shallow water (5 cm) of small creeks



Male *Pseudomugil* sp. sparring, Dekai.

Photo: Gary Lange

After this success we packed up all the fishes with fresh water in breathing bags and headed back to Dekai. Our plane was already waiting for us. The flight back to Sentani had two stopovers, one in a mountain village called Borme and another stop in the village of Gidomen, where we were able to make another short fishing trip. But that is another story. Back to Sentani, we kept the fishes in provisional aquariums until the departure for the way home. We transported the fishes to Jakarta, handed them over to a professional exporter and got them sent over to Germany just a few weeks later. That worked pretty well and all the Dekai fishes survived the transport. We were able to set up breeding groups in Germany as well as in the USA and all three Dekai Rainbowfishes and the blue-eye are well established in the hobby now.



Johannes Graf, Dan Dority and Gary Lange proudly display their catch. Photographer unknown



Male *Melanotaenia garylangei*. Photo: Gary Lange



Keeping and Raising *Melanotaenia garylangei* – The Golden Rainbowfish from Dekai

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Melanotaenia garylangei (Golden Rainbowfish) is a part of the “Maccullochi” group, so it is a medium sized rainbowfish. I feed quality flake food, freeze dried bloodworms and frozen adult brine shrimp, as well as frozen bloodworms. Sometimes for a treat they get live blackworms. Certainly if you have other live foods such as mosquito larvae or daphnia these are also A grade food choices. As per the “Ron Bowman method” my fish get one day of fasting every week. I keep the majority of my rainbowfishes in semi-planted tanks, so there may be a bit of plant nibbling but it is really not noticeable with *M. garylangei*. I avoid hairy (fine-leaved) plants because I want the fish to put their eggs in the yarn mop provided, rather than on the plants. The plants in their tank usually include *Anubias*, and *Cryptocoryne* or Swordplants, as these are easy to keep and don’t require a lot of attention.

I haven’t found the water temperature to be all that important, but I usually keep it around 24°C (75°F). I try to maintain a 50% weekly water change on all of my rainbowfish tanks, which is especially important when you are attempting to breed them. I use a floating mop that is just shorter than the height of the tank including gravel. I try not to let the ends drag on the bottom of the aquarium. My mops are usually 120 strand, which is 60 revolutions around the right sized book, tied off, and cut on the bottom. I add a styrofoam float, wash it well and it is ready to receive eggs. After seven days, if they’ve been producing eggs, I remove the mop and move it to a bare 20 or 40 litre tank (5.5 or 10 US gallons) to hatch the eggs. The eggs are incubated at 27–28°C (80–82°F) with just a slight amount of aeration.



Adult *Melanotaenia garylangei*.

Photo: Gary Lange

If I’m preparing eggs for shipping or egg vials for my talks, I pick them and place them into a shallow tray, adding a small amount of methylene blue to lightly tinge the water. Overnight incubation with methylene blue present will usually stain any eggs that are infertile, allowing them to be detected and removed. One of the advantages of picking and staining eggs is that it can break the TB (Mycobacteriosis) cycle. I’ve picked eggs from infected rainbowfish and have successfully broken this infection cycle every time. If you just move the mop of eggs you have a good chance of also moving some of the Mycobacteria to the hatching tank. That is a trick that every rainbowfish-keeper should know, but many people are surprised when I tell them about it. I guess it is the low copy number of bacteria that are actually present that helps break the cycle, nothing magical about placing eggs in methylene blue, I want to be clear on that.



Juvenile *Melanotaenia garylangei*.

Photo: Gary Lange

But back to hatching fry. Once the fry have started hatching, feed them a food small enough for them to eat. In the US that is Golden Pearls 5–50 micron size. If you can raise a lot of vinegar eels they will also work well. Micro-worms really don't work very well as they sink and the fry really only swim in the top centimeter or so of water for several days. If that is all you have to feed though that is better than them starving to death. After around five days they start feeding on live baby brine shrimp and in a few weeks will feed on larger dry prepared food. As for all rainbowfish fry, water change at least 10% a week to remove any waste at the bottom of the tank for the first month or so. Adding a few snails is a good idea to help clean up any overfeeding issues. Make sure that any water you add is the same temperature as they are, like most rainbowfish fry, quite sensitive to change.

As the fry grow larger make sure you don't crowd them, but start performing larger water changes to keep up the water quality. In about three months you should start seeing differences in the two sexes. This species starts producing viable eggs at a fairly small size – slightly over 2.5 cm in total length.



Adult *Melanotaenia garylangei*.

Photo: Gary Lange

If you get a chance to keep some of these, by all means get them. Bright colors and their moderate size make them a great rainbowfish for just about any aquarium!



Adult *Melanotaenia garylangei*.

Photo: Gary Lange

